

Visualizing the Geographic and Demographic Distribution of COVID-19

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Abstract

Whereas African Americans are disproportionately among the coronavirus disease 2019 (COVID-19) pandemic's sick and dead, less is known about whether some racial/ethnic groups are more likely to be affected in Canada. In this data visualization, the authors address two issues limiting understanding of the spatial and demographic distribution of the COVID-19 pandemic in Canada: (1) COVID-19 infection and death counts are collected at a very high level of geographic aggregation, and (2) these counts are not tallied by sociodemographic group, including race/ethnicity. The authors use a bivariate choropleth map to illustrate the correlation between COVID-19 infections and the percentage of residents who are Black across census subdivisions. Canada is more similar to the United States than expected in this respect: areas with higher shares of Black Canadians also see more infections.

Keywords

race/ethnicity, health, COVID-19, Canada, mapping

As communities, health agencies, and governments continue to deal with the coronavirus disease 2019 (COVID-19) pandemic, understanding how its impact varies over space and across population groups is crucial. But in Canada, two problems arise with such efforts. First, counts of COVID-19 cases are collected and released by Canada's health regions. These geographic areas are used by the provincial governments to administer and disperse health care resources, but they are too large and too heterogeneous to meaningfully understand the impact of the pandemic on more local communities and specific subpopulations.

Second, the COVID-19 data are not tallied separately by race/ethnicity, socioeconomic status, or other demographic groups. Black Canadians, for example, constitute about 3.5 percent of the total population, with higher shares in cities such as Montreal (6.8 percent) and Toronto (7.5 percent), but it is not clear whether they are represented at disproportionately high rates among COVID-19 infections and deaths. In contrast, African Americans' overrepresentation among those affected by COVID-19 in the United States is well documented (Thebault, Tran, and Williams 2020). There are reasons to believe that this may or may not be the case in Canada. Despite Canada's national policy of multiculturalism (Brosseau and Dewing 2013), Black Canadians experience similar disadvantages and discrimination as their American counterparts, including in the labor market (Attewell,

Kasinitz, and Dunn 2010). At the same time, Black Canadians generally report comparable or better health relative to white Canadians (Lebrun and LaVeist 2013; but see Veenstra and Patterson 2016).

In this visualization, we address both issues and provide an example of illustrating the spatial distribution of two demographic characteristics simultaneously. To examine the geographic spread of COVID-19 cases at a lower level of aggregation than the health region, we imputed COVID-19 infections for census subdivisions (akin to minor civil divisions, county divisions, or incorporated places in the United States). We excluded northern Canada (Yukon, the Northwest Territories, and Nunavut) given their sparse populations and low COVID-19 counts. We modeled the number of infections in a health region as a function of demographic factors including median after-tax income, percentage Black, percentage foreign born, percentage 65 years and older, and population size and density. We then used these results to allocate infections across each health region's constituent subdivisions (more details are provided in the Supplemental

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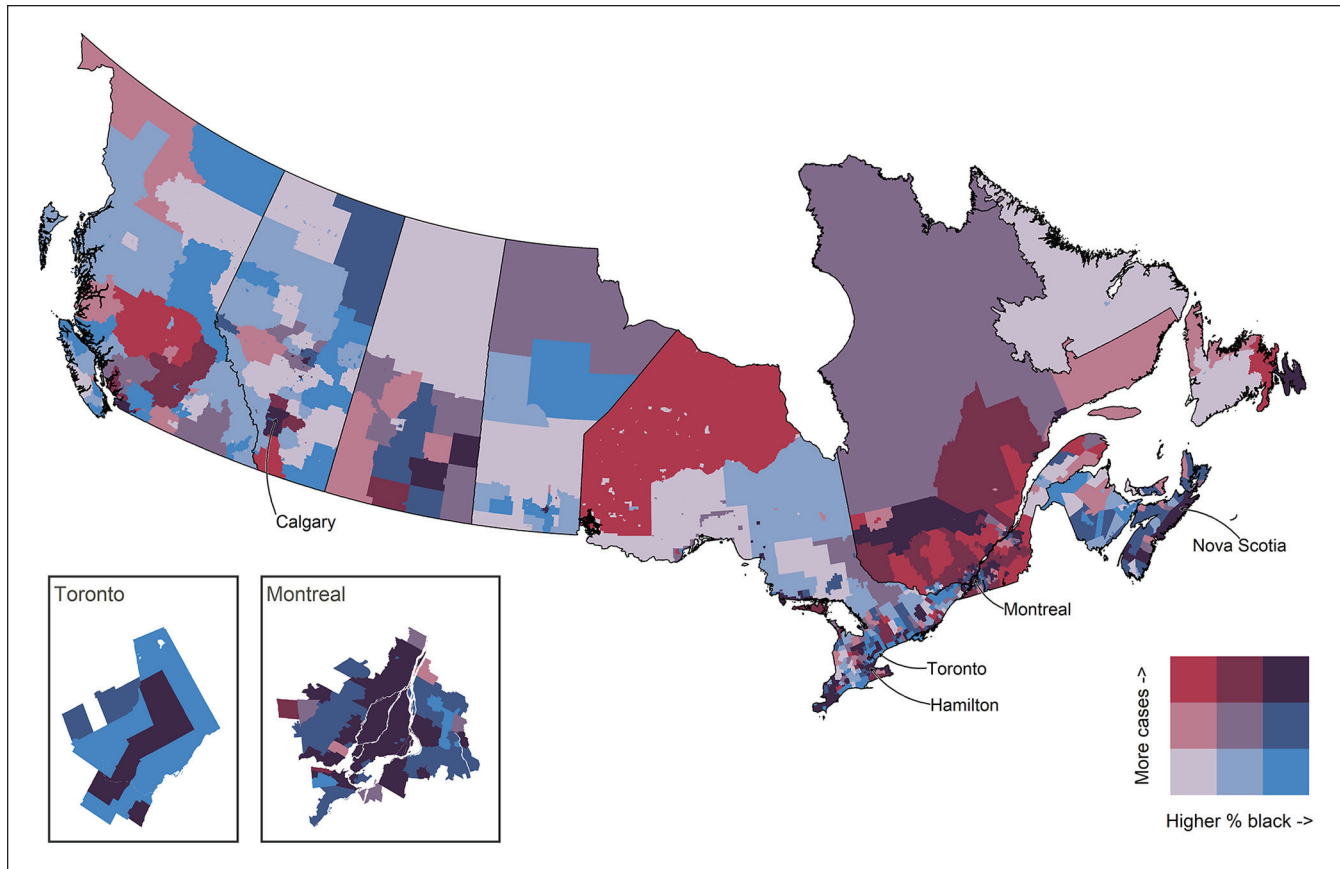


Figure 1. COVID-19 in Canada: percentage Black residents and COVID-19 infections across census subdivisions.

Note: Data are publicly available from the Public Health Agency of Canada and the 2016 Canadian census. This bivariate choropleth map illustrates the spatial distribution of the predicted count of coronavirus disease 2019 (COVID-19) infections and the percentage of residents who are Black across Canada's census subdivisions ($n = 967$). Provinces are outlined in black. The labeled cities (Calgary, Hamilton, and Montreal) and province (Nova Scotia) are discussed in the main text. Insets enlarge the census subdivisions in the Toronto and Montreal census metropolitan areas. Canada's northern provinces and territories are excluded.

Material). Absent individual-level demographic and COVID-19 data, and thus the ability to say whether Black Canadians are more or less likely to become infected by COVID-19, the choropleth map also shows where relatively high proportions of Black residents live.

Figure 1 maps the distribution of COVID-19 infections and the percentage of residents who are Black across Canada's census subdivisions.¹ As expected, there are more cases in densely populated areas such as Montreal, Canada's second largest city. Its province, Quebec, has emerged as a "hotspot" for COVID-19 cases, and although some have speculated that this is due in part to the city's and province's relatively large Black population, we provide among the first evidence that this is the case. More generally, communities in which there are higher shares of Black residents also experience higher infection counts. These two measures are positively and moderately correlated ($r = .43$).

¹Please see the Supplemental Material for a more detailed discussion about our census subdivision measure.

This is not due simply to larger population centers or higher population density: dark purple areas—communities with high percentages of Black residents and high COVID-19 counts—are found in more and less densely populated areas. For instance, large cities such as Calgary (Canada's fourth largest) and Hamilton (a city near Toronto with population of more than half a million) as well as areas in Nova Scotia (a province with fewer than 1 million people overall) have both relatively high COVID-19 counts and relatively high shares of Black residents.

This data visualization illustrates the utility of bivariate choropleth maps for showing where and which communities are most affected by the present pandemic. In addition to highlighting the surprising similarities between Canada and the United States in terms of racial inequality (i.e., the vulnerability of communities with higher shares of Black residents in both countries), the information in this kind of map may prove useful to policymakers as they continue to craft effective and targeted responses to the mass health crisis.

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Supplemental Material

Supplemental material for this article is available online.

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Author Biographies

Patrick Denice is an assistant professor of sociology at the University of Western Ontario. His research interests focus on inequality in education and the labor market. Some of his current work examines

non-normative and extended pathways to and through postsecondary education, the implications of workplace policies and institutions for workers' wages, and how public school choice policies shape patterns of racial/ethnic segregation and stratification within and across schools. His recent work has been published in *Social Science Research*, *Sociology of Education*, and *Demography*.

Kate H. Choi is an associate professor in sociology at the University of Western Ontario. Her research examines the nature, determinants, and consequences of racial and socioeconomic inequality. Within this broad theme, she is particularly interested in (1) the determinants of partner selection and consequences of marital sorting for inequality in future generations and (2) inequality by race and socioeconomic status. Her recent work has been published in *Demography*, the *Journal of Marriage and Family*, and *International Migration Review*.

Michael Haan is an associate professor and Canada Research Chair in Migration and Ethnic Relations at Western University. His research interests intersect the areas of demography, immigrant settlement, housing markets, labour markets, and data development. Some of his current work looks at the primary and secondary location choices of newcomers to Canada and the factors that shape these choices. His work has been published in a wide array of journals, including *Population*, *Space, and Place*, *International Migration Review*, and the *Journal of Marriage and Family*.

Anna Zajacova is an associate professor of sociology at the University of Western Ontario. Her background is in sociology, demography, and social epidemiology. Merging these perspectives, she studies population health, health inequalities, and social determinants of health over the adult life course. In particular, she aims to understand how experiences in secondary and higher education influence later life health and longevity among adults in the United States and Canada. Her work has been funded by the National Institute on Aging and the National Cancer Institute and published in outlets such as *Demography*, *Social Science and Medicine*, and the *Journal of Pain*.